Coordinate Geometry

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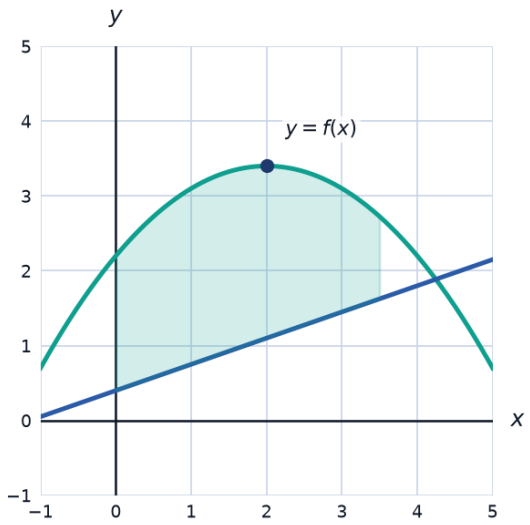
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the diagram below, two chords, AB and CD, intersect at point P inside a circle. If $AP = 5$ cm, $PB = 8$ cm, and $CP = 4$ cm, what is the length of PD in cm?

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the figure below with $AB \parallel CD$, $\text{Area}(\Delta ABE) = 36$ and $\text{Area}(\Delta CDE) = 81$. If base $AB = 10$, what is the length of base CD?



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The triangle ABC is translated on the coordinate plane according to the rule $(x, y) \to (x - 3, y + 2)$. If the vertices of the image triangle A'B'C' are A'(-1, 5), B'(3, 1), and C'(-2, -3), what are the coordinates of the original vertices A, B, and C?

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In ΔABC, $AD = 3$ and $DB = 9$ (so $AB = 12$); $AE = 4$ and $EC = 5$ (so $AC = 9$). If base $BC = 21$, what is the length of segment DE?

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The diagonals of trapezoid $ABCD$ with $AB \parallel CD$ intersect at P. If $\text{Area}(\Delta APB) = 16$ and $\text{Area}(\Delta CPD) = 49$, what is the total area of trapezoid $ABCD$?

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In ΔABC, segment $DE \parallel BC$ with D on AB and E on AC. If $AD = 5$, $DB = 10$, and $DE = 7$, what is the length of base BC?

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7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the hourglass figure below, segment $AB \parallel CD$, and segments AC and BD intersect at E. If $AB = 8$, $CD = 12$, and $AE = 6$, what is the length of segment CE?

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A right triangle has legs of lengths 15 and 20. What is the exact length of the altitude drawn perpendicular to the hypotenuse?

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In ΔABC, $\angle ADE \cong \angle C$ with $AD = 5$ and $AC = 15$. If the total area of ΔABC is 108, what is the area of quadrilateral $BDEC$?

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In ΔABC shown below, segment $DE \parallel BC$. If $AD = 6$, $DB = 9$, and $AE = 8$, what is the length of segment EC?

11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In ΔABC, $DE \parallel BC$ with D on AB and E on AC. If $AD = 4$ and $AB = 10$, what fraction of the area of ΔABC is occupied by trapezoid $BDEC$?

12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Trapezoid $ABCD$ has parallel bases $AB = 6$ and $CD = 14$. The diagonals AC and BD intersect at point P. What is the ratio of $\text{Area}(\Delta APB)$ to $\text{Area}(\Delta CPD)$?

13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In ΔABC, point D lies on AB and E on AC such that $\angle AED \cong \angle B$. If $AE = 6$, $AD = 8$, $AC = 18$, and base $BC = 27$, what is the length of segment DE?

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In trapezoid $ABCD$ with $AB \parallel CD$, $AB = 8$ and $CD = 12$. The diagonals intersect at P. If the entire diagonal $AC = 30$, what is the length of segment AP?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Right Triangles & Trigonometry

In the bowtie figure with $AB \parallel CD$, segments AC and BD intersect at E. If $AB = 10$, $CD = 15$, and the entire segment $BD = 35$, what is the length of segment BE?

16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Polygons & Quadrilaterals

In ΔABC, ray AD bisects $\angle A$ with point D on side BC. If $AB = 8$, $AC = 12$, and $BD = 6$, what is the length of segment DC?

17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Circles

In ΔABC, ray AD bisects $\angle A$. If $AB = x$, $AC = x + 4$, $BD = 4$, and $DC = 6$, what is the length of side AC?

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Area, Surface Area & Volume

In right ΔABC, altitude CD to hypotenuse AB measures 6, and the entire hypotenuse $AB = 13$. What is the length of the longer of the two segments into which D divides AB?

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Transformations & Symmetry

In ΔABC, segment $DE \parallel BC$. If $AD = 8$, $DB = 4$, and $AC = 18$, what is the length of segment AE?

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Foundations & Reasoning

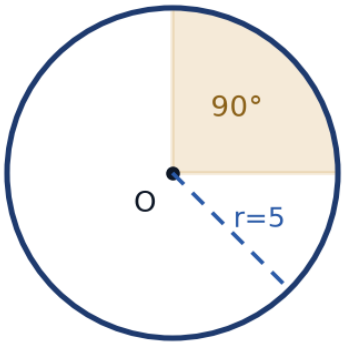
In ΔABC, ray AD bisects $\angle A$ with D on BC. If $AB = 10$, $AC = 15$, and the entire base $BC = 20$, what is the length of segment BD?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 6x + 6$ in review: circumference and area, evaluate $f(2)$.



a. 18

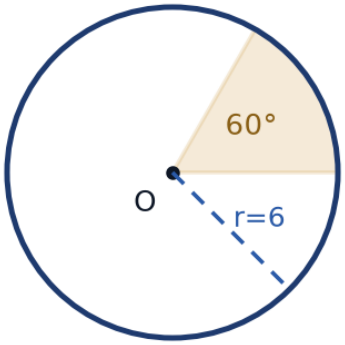
b. 24

c. 12

d. 22

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 7$ in arc length and sector area, evaluate $f(2)$.



a. 11

b. 13

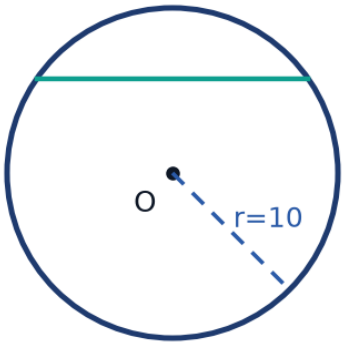
c. 4

d. 15

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 8$ in extension: chords of a circle, evaluate $f(2)$.



a. 14

b. 17

c. 6

d. 18

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for extension: chords of a circle consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5

d. 25.5

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In ΔABC studying angles from intersecting chords, secants, and tangents, $m\angle A = 43^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 85°

b. 95°

c. 77°

d. 137°

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for tangents to a circle consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

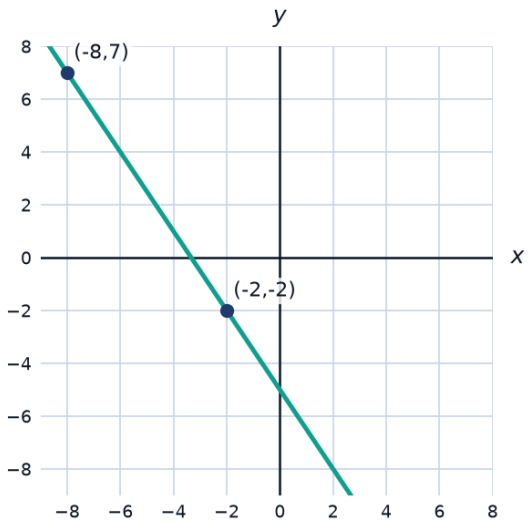
b. 16.5

c. 17.5

d. 19.5

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the midpoint of the segment connecting $(-2, -2)$ and $(-8, 7)$.



a. (-6, 9)

b. (-5, -2)

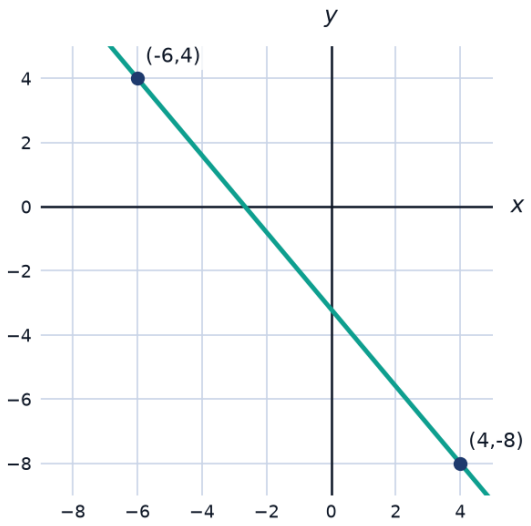
c. (-10, 5)

d. (-5, 5/2)

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28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the distance between the points $(-6, 4)$ and $(4, -8)$.



a. $3\sqrt{5}$

b. $2\sqrt{61}$

c. 22

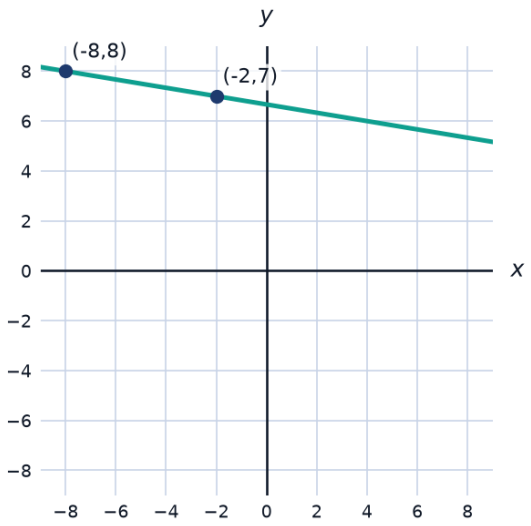
d. $2\sqrt{62}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the slope of the line through $(-8, 8)$ and $(-2, 7)$.



a. 1/6

b. -1/6

c. -6

d. 6

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In the coordinate plane, quadrilateral $ABCD$ has vertices $A(-2, 3)$, $B(4, 5)$, $C(6, -1)$, and $D(0, -3)$. Which of the following correctly classifies quadrilateral $ABCD$ and gives its exact area?

a. A square with an area of 40

b. A rectangle that is not a square, with an area of 40

c. A rhombus that is not a square, with an area of 40

d. A square with an area of 64

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the xy-plane, a parabola $\mathcal{P}$ is defined as the set of all points equidistant from the focus $F(1, 2)$ and the directrix line $y = -2$. The line $x - y = 1$ intersects $\mathcal{P}$ at two distinct points, A and B. What is the length of the segment AB?

a. $4\sqrt{2}$

b. 8

c. $8\sqrt{2}$

d. 16

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Which of the following is an equation of the line that passes through the point $(4, -1)$ and is perpendicular to the line $y = 2x + 3$?

a. $y = -\frac{1}{2}x + 1$

b. $y = -\frac{1}{2}x - 3$

c. $y = 2x - 9$

d. $y = \frac{1}{2}x - 3$

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the equation (in slope-intercept form) of the line through $(-8, -8)$ with slope $1/2$.

a. y=1/2x-4

b. y=1/2(x-(-8))

c. y=1/2x

d. y=-4x+1/2

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A line has slope -3. What is the slope of a line perpendicular to it?

a. 1

b. 1/3

c. -3

d. 3

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In a plane, two horizontal lines ℓ_1 and ℓ_2 are parallel, with ℓ_1 positioned above ℓ_2. A zig-zag transversal path connects point A on ℓ_1 to point D on ℓ_2 through intermediate vertices B and C, which lie strictly between the two parallel lines. - Segment AB extends downward and to the right from A, forming an acute angle of 65° with line ℓ_1. - Segment BC extends downward and to the left from B to C. - Segment CD extends downward and to the right from C, meeting line ℓ_2 at an acute angle of 25°. If the ray that bisects $\angle BCD$ is perpendicular to segment AB, what is the measure of $\angle ABC$?

a. 115°

b. 130°

c. 140°

d. 150°

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A quadrilateral in the coordinate plane has vertices at $(0, 0)$, $(3, 0)$, $(3, 5)$, and $(0, 9)$. What is the volume of the three-dimensional solid generated by rotating this region 360° about the y-axis?

a. 57π

b. 66π

c. 72π

d. 81π

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the coordinate plane, point $P(7, -2)$ is rotated 90° counterclockwise about the point $C(3, 1)$ to produce image point P'. What are the coordinates of P'?

a. $(6, 5)$

b. $(3, 4)$

c. $(2, 7)$

d. $(0, -3)$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Area, Surface Area & Volume

A right circular cone with radius R and height H is sliced by a plane parallel to its base at a distance of $\frac{1}{3}H$ from the apex. What is the area of the 2D cross-section formed by this plane?

a. $\frac{1}{9}\pi R^2$

b. $\frac{1}{3}\pi R^2$

c. $\frac{4}{9}\pi R^2$

d. $\frac{2}{3}\pi R^2$

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 9.01 Transformations & Symmetry

In the coordinate plane, ΔABC has vertices $A(0, 0)$, $B(4, 0)$, and $C(2, 4)$. A dilation centered at the origin maps ΔABC to ΔDEF with vertices $D(0, 0)$, $E(10, 0)$, and $F(5, 10)$. If the area of ΔABC is 8, what is the area of ΔDEF?

a. 50

b. 20

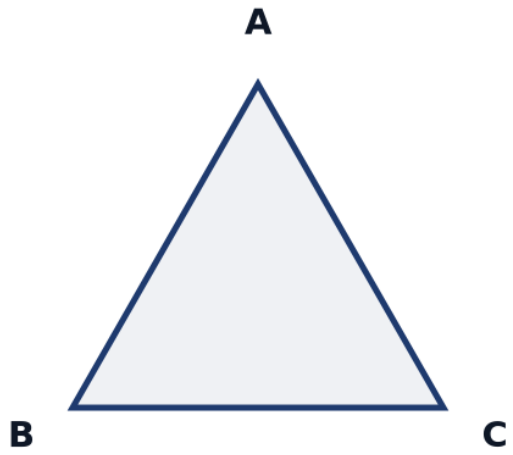
c. 40

d. 64

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Foundations & Reasoning

In ΔABC, side AB is congruent to side AC, and the measure of $\angle A$ is 50°. What is the measure of $\angle B$?



a. 50°

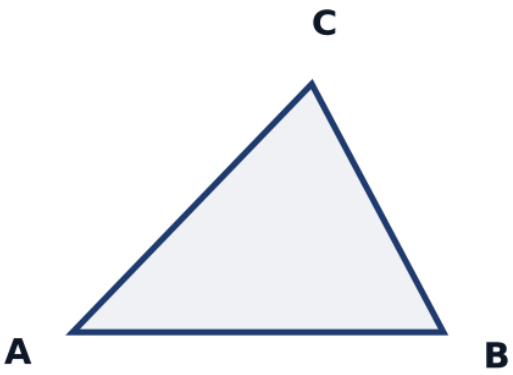
b. 65°

c. 80°

d. 130°

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 11.01 Probability

Triangle ABC has $\angle A = 46^\circ$ and $\angle B = 62^\circ$. A line through C parallel to AB is drawn, forming an exterior angle at C on the far side from the triangle. What is the measure of that exterior angle, and how does it relate to $\angle A$ and $\angle B$?



a. 72°

b. 108°

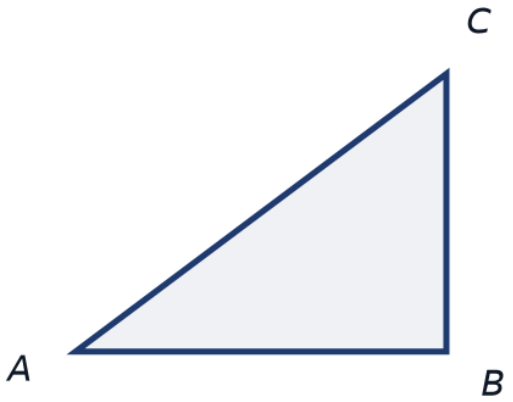
c. 46°

d. 62°

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Lines & Angles

Pythagorean theorem applies to:



a. All triangles

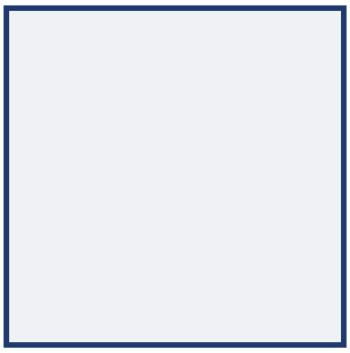
b. Equilateral

c. Right triangles

d. Obtuse

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Triangles & Congruence

A point is chosen at random inside a 19×19 square. A smaller 15×15 square sits in one corner. Find the probability the point lands inside the smaller square.



a. $\frac{15}{19}$

b. $\frac{225}{362}$

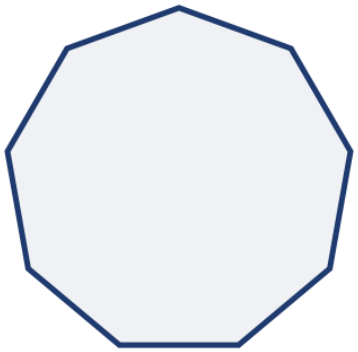
c. $\frac{136}{361}$

d. $\frac{225}{361}$

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

A regular 9-gon has each interior angle equal to $(4x + 28)^\circ$. Find the value of the interior angle.



a. 28°

b. 155°

c. 140°

d. 1260°

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Right Triangles & Trigonometry

Trapezoid parallel sides 5 and 9 height 4: area:



a. 14

b. 24

c. 28

d. 56

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